

PROTOCOL v1.0 | 5 AUGUST 2026 | EXECUTION BEFORE 12 AUGUST 2026

Experimental Performance Evaluation of the Unitree L1 3D LiDAR

Accelerated and reproducible protocol for a short paper in indoor mobile robotics

MINIMUM SCIENTIFIC OBJECTIVE

Before departure, produce a clean dataset and a defensible analysis of (1) planar-target range accuracy and precision, (2) coverage convergence of the non-repetitive scan, (3) ROS 2 stream stability, and (4) indoor mapping repeatability. The study must not claim absolute SLAM accuracy without ground truth.

Prepared for

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1. Experimental scope and decision

The available time requires a clear separation between the core contribution and optional extensions. The core contribution must remain a simple, quantitative, and repeatable instrumental characterisation. Mapping is used to demonstrate application-level impact, not as evidence of absolute accuracy.

Level	Work package	Status before departure
Core A	Planar-target range: bias, repeatability, point-to-plane noise, valid returns	Mandatory
Core B	Angular profile and coverage convergence of the non-repetitive scan	Mandatory
Core C	ROS 2 pipeline stability over 30 minutes	Mandatory
Application	Three identical mapping traversals and relative comparison	Strongly recommended
Extension	Co-located comparison with an Ouster-64 or another reference sensor	Only if immediately available

SUCCESS CRITERION

A negative result can be publishable when it is measured correctly. An attractive map without a protocol, repetitions, metadata, or uncertainty is not.

2. Research question and hypotheses

Main research question. Under which conditions does the Unitree L1 provide measurements and spatial coverage that are sufficiently stable for perception and mapping on an indoor mobile robot?

- **H1 - range.** For a matte, frontal, well-illuminated target, the mean bias is expected to remain compatible with the manufacturer-stated ± 2 cm accuracy, while dispersion increases with range and incidence angle.
- **H2 - scanning pattern.** Angular density is non-uniform and coverage improves with accumulation time; the paper should therefore report a coverage-versus-time curve rather than only a final accumulated point cloud.
- **H3 - software pipeline.** Useful throughput, timing jitter, and the invalid-point rate can be characterised independently of the SLAM algorithm.
- **H4 - mapping.** Repeated traversals should produce globally consistent geometry at navigation scale, even if fine reconstruction remains limited by point density and scan overlap.

3. Instrument specifications to verify before acquisition

The manual distinguishes two variants, L1 PM and L1 RM. The exact unit available in the laboratory must be photographed and recorded in the experiment manifest. The values below are manufacturer specifications, not experimental results [1].

Parameter	Stated specification	Implication for the protocol
Variant	L1 PM (20 m at 90% reflectivity) or L1 RM (30 m)	Identify the exact model; do not mix the range limits.
Measurement accuracy	±2.0 cm	Test planar-target bias using an independent reference.
Resolution	8 mm	Do not confuse numerical resolution with accuracy.
Field of view	360° × 90°, hemispherical	Describe the physical sensor orientation.
Effective point rate	21,600 points/s	Measure the rate actually received by ROS 2.
Scan pattern	Non-repetitive; higher density near the centre	Measure coverage as a function of time and direction.
IMU	Reported at 250 Hz	Check frequency, timestamp monotonicity, and saturation.
Near blind zone	0.05 m	Keep mounts and cables outside the useful FOV.

4. Preparation and configuration freeze

1. Photograph the mounting arrangement, sensor label, serial number, and axis orientation.
2. Clean the optical window and allow the sensor to warm up for 10 minutes before the first acquisition.
3. Mount the LiDAR rigidly. Record the position of the sensor-frame origin and any offset relative to the point used to measure the reference distance.
4. Freeze the software configuration: clean Git repository, identified commit, Docker image or package list, YAML files, static transforms, and launch commands.
5. Verify the topics that are actually published. For the current setup, check at least /unilidar/cloud, /unilidar/imu, /tf, and /tf_static; add Jackal odometry for mobile experiments.
6. Create a CSV experiment manifest before recording the first rosbag. No data file should exist without a corresponding metadata row.

GO / NO-GO CRITERION

Start the test matrix only when the monitoring node confirms monotonic timestamps, a known frame, no persistent driver error, and successful reopening of a test rosbag.

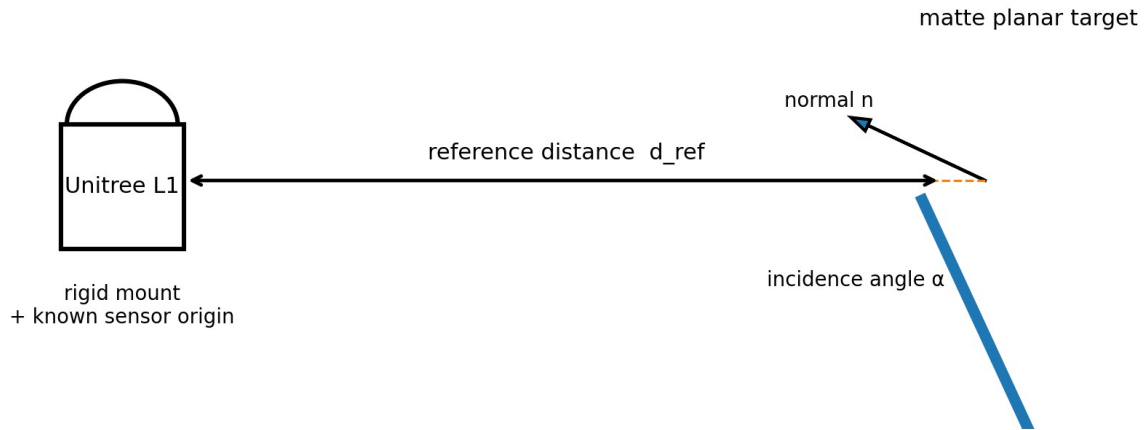


Figure 1 - Minimum geometry for the planar-target experiment. The reference distance must be measured from the LiDAR frame origin or corrected using a measured offset.

5. Experiment A - range accuracy and precision

5.1 Minimum equipment

- Matte planar target at least 0.8 m × 0.8 m, ideally white; avoid warped edges.
- Verified laser distance meter or steel tape measure. Record the resolution and uncertainty of the reference instrument.
- Rigid LiDAR mount and an adjustable support for the target.
- Masking tape or floor markers to reproduce the positions.
- Room with no moving people, no window directly in the measurement field, and stable illumination.

5.2 Feasible test matrix

Series	Variable	Levels	Repetitions	Duration per repetition
A1 - range	Range, white target, incidence close to 0°	0.5; 1; 2; 3; 5 m; 8 or 10 m if possible	3	20 s
A2 - incidence	Angle, range fixed at 2 or 3 m	0°; 30°; 60°	3	20 s
A3 - reflectance	Material, 2 m range, 0° incidence	matte white; matte black (minimum)	3	20 s

Priority: if time is limited, retain the complete A1 and A2 series and only two materials for A3. Never reduce a condition to a single acquisition.

5.3 Procedure for each acquisition

1. Place the target centre at a height compatible with the useful LiDAR field and orient the target at the planned angle.
2. Measure d_{ref} at least twice. If the two measurements differ by more than 3 mm, reposition and measure again.
3. Wait 5 seconds after any adjustment to avoid vibration and residual motion.
4. Record 20 seconds of raw point-cloud and IMU data using a standardised filename.
5. Immediately record the range, angle, material, time, approximate temperature, software commit, and any anomaly in the manifest.

6. Reopen the bag and verify its duration, message counts, and topic availability before moving the setup.

5.4 Data processing

For each acquisition, work in the LiDAR frame and extract a central region of interest on the target. Exclude the edges to limit mixed returns. First fit a robust plane with RANSAC, then re-estimate its parameters by SVD using the inliers. Reference methods commonly use point-to-plane residuals to characterise geometric noise [2-4].

Metric	Practical definition	Interpretation
Estimated distance d_{hat}	Absolute distance from the origin to the fitted plane	Robust measurement of target position.
Signed error e	$e = d_{\text{hat}} - d_{\text{ref}}$	Positive bias: target estimated too far away.
Bias per condition	Mean of the three signed errors	Systematic accuracy for the condition.
Repeatability	Standard deviation of d_{hat} across repetitions	Stability across independent acquisitions.
Point-to-plane RMSE	Root mean square of the residuals	Local noise and residual non-planarity.
Point-to-plane MAD	Median absolute deviation from the median	Dispersion robust to outliers.
Valid-return rate	Finite points in the ROI / expected or received points	Robustness of returns from the target.
Intensity / grayscale	Median and interquartile range	Radiometric indicator to be interpreted cautiously.

REFERENCE UNCERTAINTY

Report the resolution of the distance meter or tape, the uncertainty in the geometric offset between the measured point and LiDAR origin, and positioning repeatability. A claim of verification at ± 2 cm is credible only when the combined reference uncertainty is clearly below 2 cm.

6. Experiment B - non-repetitive scanning and coverage

The manual indicates direction-dependent point density, and recent work shows that non-repetitive scans may reduce scan-to-scan overlap and complicate odometry [1,5]. A simple and useful contribution is to measure how long the sensor requires to cover the observed angular space.

6.1 Acquisition

1. Place the LiDAR motionless near the centre of a clear, structured room with visible walls and ceiling.
2. Record 60 seconds of /unilidar/cloud with no motion in the scene.
3. Repeat three times, restarting the recording between acquisitions. Do not change the orientation.
4. Useful option: repeat one series with the sensor tilted by 30° to document the effect of installation orientation.

6.2 Angular analysis

- **Conversion.** For every valid point, compute azimuth $\theta = \text{atan2}(y, x)$ and elevation $\varphi = \text{atan2}(z, \sqrt{x^2 + y^2})$.

- **Binning.** Create a $2^\circ \times 2^\circ$ angular grid, then check robustness using a $1^\circ \times 1^\circ$ grid.
- **Windows.** Accumulate points over 0.1, 0.25, 0.5, 1, 2, 5, 10, and 30 s.
- **Coverage.** Ratio of occupied angular cells to observable cells in the reference scene.
- **Uniformity.** Coefficient of variation of hit counts per cell, supplemented by a count heat map.
- **Convergence.** Time required to reach 50%, 90%, and 95% of the final 30 s coverage.

Limitation to report: measured coverage depends on room geometry and valid returns. It therefore characterises the system in a specific indoor scene, not the theoretical path of every emitted ray.

7. Experiment C - ROS 2 pipeline stability

1. Start the LiDAR and monitoring node using the final configuration.
2. Record one continuous 30-minute static sequence.
3. Sample CPU and memory usage for the driver, container, and monitoring node every 5 seconds.
4. Retain the complete logs and record any human intervention or restart.

Indicator	Calculation	Report
Point-cloud message frequency	1 / mean reception interval	mean, standard deviation, minimum, maximum
Useful point rate	sum of valid points / duration	received points/s versus stated 21,600 points/s
Jitter	standard deviation and P95 of intervals	spikes, pauses, and periodicities
Timestamps	successive differences and comparison with reception time	backward step, duplicate, jump
Invalid points	NaN, Inf, out-of-range distance	overall percentage and time evolution
IMU	frequency, gaps, extreme values	comparison with stated 250 Hz
Availability	uninterrupted duration / total duration	driver restarts and dropouts
Resources	CPU and RAM	mean, P95, and drift over time

8. Experiment D - mapping repeatability

CORRECT SCIENTIFIC INTERPRETATION

Without a motion-capture system, total station, or calibrated reference sensor, this section measures relative repeatability and geometric consistency - not absolute trajectory accuracy.

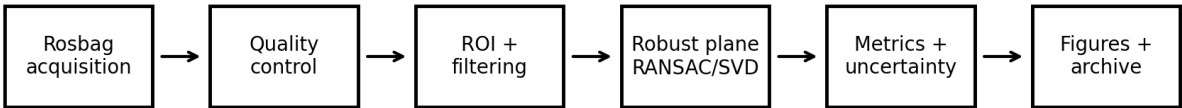
8.1 Trajectory

- Select a 20-40 m indoor loop containing a corridor, a 90° turn, doors, and a return to the starting point.
- Physically mark the start, travel direction, and two or three waypoints.
- Perform three traversals using the same configuration, a target speed of 0.2-0.3 m/s, and an environment that is as static as possible.
- Record LiDAR, IMU, TF, wheel odometry, velocity commands, and mapping-algorithm output.
- Do not change any parameter between traversals; any exception must be treated as a new experimental condition.

8.2 Relative metrics

Metric	Method	Caution
Loop-closure error	Distance and orientation between the estimated final and initial poses	Depends on the algorithm and published frame.
Cloud-to-cloud distance	Align two maps using a rigid transform, then compute median and P95 distances	Report the alignment method and compared region.
Wall thickness	Transverse standard deviation over several planar segments	Avoid doors, corners, and moving objects.
Dimensional repeatability	Measure 5-10 fixed distances in each map	Use exactly the same landmarks.
Success rate	Traversals completed without divergence or reset	Define divergence before the experiment.

High-value extension: if the Ouster-64 is immediately available, rigidly co-mount both sensors, record one trajectory simultaneously, and compare the maps after extrinsic calibration. This extension must not delay Experiments A-C.



Record every parameter change in the experiment manifest.

Figure 2 - Analysis pipeline. Raw data remain immutable; all filtering is applied by version-controlled scripts.

9. Data organisation and reproducibility

9.1 Recommended directory structure

```
unitree_l1_short_paper/  
  README.md  
  metadata/experiment_manifest.csv  
  config/driver.yaml  tf_static.yaml  software_versions.txt  
  bags/raw/  
  analysis/01_qc.py  02_plane_fit.py  03_angular_coverage.py  04_mapping_repeatability.py  
  results/tables/  results/figures/  
  paper/main.tex_or_docx  paper/references.bib
```

9.2 Naming convention

YYYYMMDD_test_variable_level_runNN

Examples: 20260807_A1_distance_2m_run01 | 20260810_D_mapping_loop_run03

9.3 Minimum manifest fields

Category	Fields
Identity	bag_name, date_time, operator, sensor_model, serial_number

Category	Fields
Geometry	distance_ref_m, incidence_deg, material, sensor_orientation, target_size
Software	git_commit, docker_image, ROS_distro, driver_parameters, mapping_parameters
Acquisition	topics, duration_s, warmup_min, run_number, environment
Quality	bag_reopened, missing_topics, anomalies, notes
Traceability	raw_file_sha256, analysis_commit, figure_version

10. Figures and tables required for the short paper

Item	Minimum content	Target status by 12 August
Figure 1	Photograph or diagram of the setup and definition of d_{ref} and α	Ready
Figure 2	Range error with repeatability bars versus reference range	Ready
Figure 3	Point-to-plane RMSE and return rate versus angle / material	Ready
Figure 4	Angular heat map + coverage-versus-time curve	Ready
Figure 5	Three maps or cloud-to-cloud comparison across traversals	Preliminary acceptable
Table 1	Sensor, platform, versions, and parameters	Ready
Table 2	Bias, repeatability, RMSE, and valid returns by condition	Ready
Table 3	Frequencies, jitter, throughput, invalid points, CPU/RAM	Ready

11. Structure of the short manuscript

Section	Main message	Indicative length
Abstract	Question, method, main quantitative results, limitation	150-200 words
1. Introduction	Need for independent characterisation; non-repetitive scanning; contributions	0.5 page

Section	Main message	Indicative length
2. Sensor and platform	L1 PM/RM, Jackal, ROS 2, setup, software	0.5 page
3. Method	A-C mandatory, D secondary, repetitions and metrics	1-1.5 pages
4. Results	Plots, intervals, failures, and observations	1-1.5 pages
5. Discussion	Favourable conditions, limitations, navigation-versus-reconstruction impact	0.5-1 page
6. Conclusion	Cautious response to the hypotheses and open data if authorised	0.25 page

CLAIM THIS CONTRIBUTION ONLY IF SUPPORTED BY THE DATA

"We present a reproducible characterisation of Unitree L1 range bias, point-to-plane noise, time-dependent angular coverage, and ROS 2 stability, complemented by an indoor mapping repeatability study."

12. Critical schedule: 6-12 August 2026

Date	Main work block	End-of-day deliverable
6 August	Supervisor validation; PM/RM model; code freeze; complete pilot	One pilot rosbag analysed end to end and a validated manifest
7 August	Experiment A1 range + immediate quality control	15 range acquisitions with no corrupted bag
8 August	A2 incidence + A3 materials + missing repetitions	Complete static test matrix
9 August	Experiment B coverage + C 30-minute stability	Three coverage series and one long-duration test
10 August	Three D trajectories and repetition of doubtful tests	Three comparable mobile bags
11 August	Scripts, tables, figures, backups, and cross-checking	Figures 1-4 and Tables 1-3
12 August	Presentation to supervisors; clean repository; manuscript skeleton; follow-up task allocation	Archive transferred, responsibilities assigned, and next meeting scheduled

13. Quality control and research integrity

- **Pre-register exclusions.** Define before analysis what invalidates a run: motion, incomplete bag, target partly outside the ROI, or a configuration change.
- **Preserve raw data.** Never overwrite rosbags. Filtered or downsampled files belong in a derived-data directory.

- **Show variability.** Report repetitions, error bars, and sample counts; do not present only the best acquisition.
- **Separate bias from noise.** Mean error and point-to-plane dispersion answer different questions.
- **Limit the claims.** Do not present relative repeatability as absolute accuracy, or a manufacturer specification as a measured result.
- **Authorship and ownership.** Confirm with Prof. Nunes and Dr Barros the author order, data ownership, code release, and any public distribution of the rosbags.

14. Printable field sheet

Check before each run	OK / value
Sensor and variant photographed; optical window clean	
Warm-up ≥ 10 min before the first run	
Rigid mounting; orientation and TF unchanged	
d_ref and angle measured; origin offset accounted for	
Bag name compliant and manifest row created	
Cloud / IMU / TF topics present	
No person or moving object in the ROI	
Bag reopened after acquisition; duration and messages verified	
Anomalies recorded immediately	

Decision: ☐ VALID RUN ☐ REPEAT REQUIRED **Reason:** _____

15. Methodological references

- [1] Unitree Robotics, Unitree 4D LiDAR-L1 User Manual, version 1.1, June 2024. Specifications: PM/RM range, ± 2 cm stated accuracy, $360^\circ \times 90^\circ$ FOV, 21,600 effective points/s, non-uniform point density.
- [2] B. Muralikrishnan, "Performance Evaluation of Terrestrial Laser Scanners—A Review," *Measurement Science and Technology*, vol. 32, no. 7, 072001, 2021. DOI: 10.1088/1361-6501/abdae3.
- [3] S. Soudarissanane, R. Lindenbergh, M. Menenti, and P. J. G. Teunissen, "Incidence Angle Influence on the Quality of Terrestrial Laser Scanning Points," *ISPRS Workshop Laser Scanning*, 2009.
- [4] J. Song and C. Harmening, "Quantitative Evaluation of Angular, Range, and Reflectance Influences on Laser Scanners Using a High-Precision Linear Rail Test System," *ISPRS Archives*, vol. XLVIII-1/W6-2025, pp. 219–225, 2025. DOI: 10.5194/isprs-archives-XLVIII-1-W6-2025-219-2025.
- [5] P. Habibiroudkenar et al., "Non-Repetitive Scan Correction for LiDAR Odometry via Spatial Analysis in CERN Facilities," *IEEE Access*, vol. 14, pp. 52717–52733, 2026. DOI: 10.1109/ACCESS.2026.3680632.
- [6] A. Hidding et al., "Robotic Lava Tube Mapping and Multimodal Data Collection Using Quadraped and LiDAR," *Frontiers in Robotics and AI*, 2026. DOI: 10.3389/frobt.2026.1855443.

MINIMUM DELIVERABLE BEFORE LEAVING COIMBRA

Before departure: archive with metadata and checksums; four version-controlled scripts; Figures 1-4; Tables 1-3; manuscript skeleton; validation meeting; list of remaining tasks.